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Lab Report

Phishing Simulation & Campaign Tracking (Gophish)

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1. Executive Summary

This report documents a hands-on lab exercise using Gophish, a professional open-source phishing simulation framework. Gophish provides a full-featured web-based dashboard for managing phishing campaigns — including sending profiles, email templates, landing pages, and target groups — making it a standard tool in authorized red-team and security awareness exercises.

The lab walkthrough covers the complete Gophish campaign lifecycle from installation to campaign launch:

- Installation and first-time login with password reset.
- Configuring a Sending Profile using Gmail SMTP with an App Password for authenticated email delivery.
- Verifying the sending profile by dispatching a test email and confirming receipt.
- Creating a Landing Page by importing a real website, enabling credential capture and password logging.
- Building an Email Template by importing a legitimate real-world email (Smallpdf) as a convincing lure.
- Creating a Users & Groups target list with multiple recipients.
- Launching a Campaign that ties together the template, landing page, URL, sending profile, and target group.
- Reviewing campaign delivery status and result details.

> **Important:** Gophish and all phishing simulation tools must only be used in authorized environments with explicit written consent. Unauthorized phishing is illegal under computer fraud and anti-abuse laws in all jurisdictions.

2. Introduction to Gophish

Gophish is a free, open-source phishing toolkit written in Go. It is designed specifically for penetration testers and security teams to rapidly deploy and track phishing campaigns. Its browser-based admin panel simplifies every stage of a campaign — from email template creation to real-time click/submission tracking.



2.1 Tool Overview

Tool Name	Gophish
Default User	admin
Default Pass	kali-gophish (reset on first login)
Platform	Kali Linux (also: Windows, macOS, Linux)
SMTP Support	Gmail, custom SMTP, open relays

2.2 Campaign Workflow Overview

A Gophish campaign requires five components to be configured before launch. Each must be set up in the dashboard before creating a campaign:

#	Component	Purpose
1	**Sending Profile**	SMTP credentials used to send phishing emails (Gmail + App Password in this lab).
2	**Landing Page**	The fake page victims land on — captures submitted credentials.
3	**Email Template**	The phishing email content shown to targets — imported from a real email for realism.
4	**Users & Groups**	The list of target email addresses to receive the phishing campaign.
5	**Campaign**	Ties all four components together with a tracking URL, launch date, and schedule.

3. Installation & First Login

3.1 Installing Gophish

Gophish is available in the Kali Linux package repository. It can be installed with a single apt command. The package manager handles all dependencies automatically.

Step 1: Run: sudo apt install gophish — enter the sudo password when prompted.

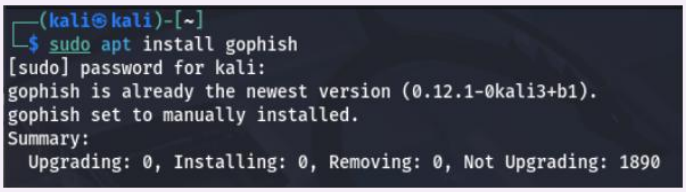


Fig 3.1 — Terminal: sudo apt install gophish

3.2 Starting Gophish

Gophish is launched from the terminal by simply running the gophish command. On startup it prints the default admin credentials and the URL for the web interface.

Step 2: Run: gophish in the terminal. Note the default username, password, and Web UI URL.

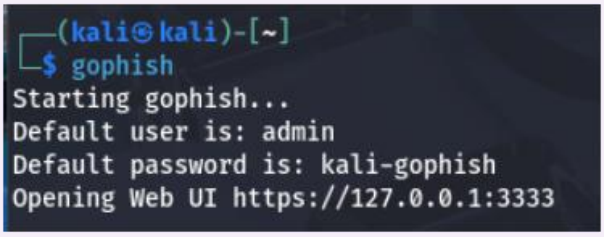


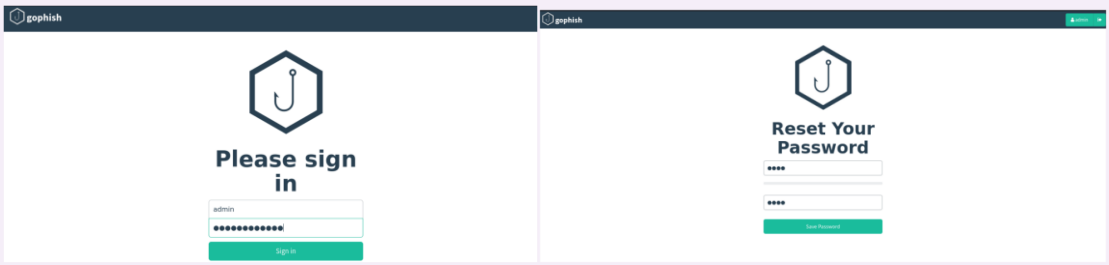
Fig 3.2 — Terminal: gophish starting

Note: Gophish generates a self-signed TLS certificate for its admin panel. Your browser will show a certificate warning — this is expected. Accept the exception to proceed to the login page.

3.3 Logging In

Open a browser and navigate to https://127.0.0.1:3333. The Gophish login form is displayed. Enter the default credentials: username admin, password kali-gophish.

Step 3: Navigate to https://127.0.0.1:3333 in the browser. Log in with admin / kali-gophish.



Gophish login: admin username and password entered, ready to sign in

Gophish web UI: login page displayed in browser

3.4 Resetting the Password

On first login, Gophish immediately redirects to a password reset page. A new secure password must be set before the dashboard becomes accessible. In this lab the password was changed to Kali1212.

Step 4: Enter and confirm a new password on the Reset Your Password screen, then save.

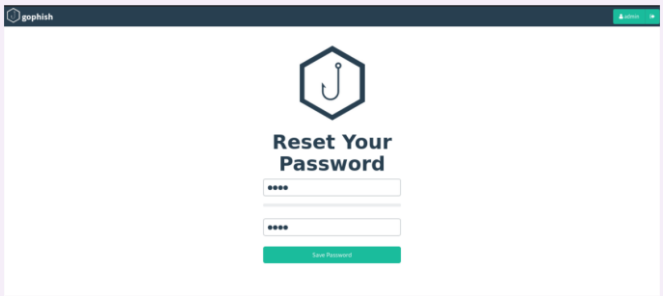


Fig 3.5 — Gophish: Reset Your Password prompt on first login (new password: Kali1212)

3.5 Dashboard Overview

After the password reset, the main Gophish dashboard is displayed. The left sidebar provides navigation to all campaign components: Campaigns, Users & Groups, Email Templates, Landing Pages, Sending Profiles, and Account Settings.

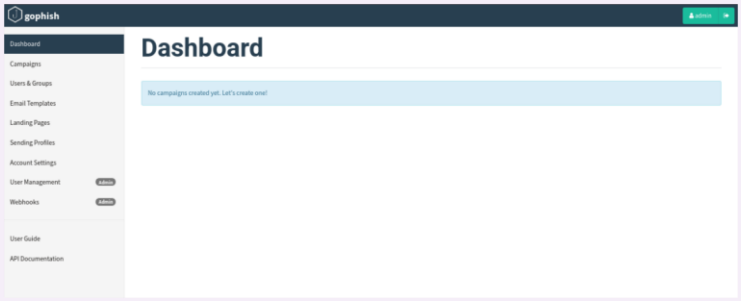


Fig 3.6 — Gophish Dashboard: empty state — 'No campaigns created yet. Let's create one!'

4. Sending Profile Configuration

A Sending Profile defines the email server and credentials Gophish uses to dispatch phishing emails. In this lab, Gmail's SMTP server was used with an App Password — a 16-character token generated by Google specifically for third-party app access, required when 2-Factor Authentication is enabled.

4.1 Creating the Sending Profile

In the Gophish sidebar, navigate to Sending Profiles and click + New Profile. Fill in the profile name, SMTP host, sender email address, and authentication credentials.

Step 5: Click Sending Profiles in the sidebar, then click + New Profile.

Step 6: Enter the following details: Name: Test-Profile | Interface Type: SMTP | SMTP From: fakefoxacc022@gmail.com | Host: smtp.gmail.com:587 | Username: fakefoxacc022@gmail.com.

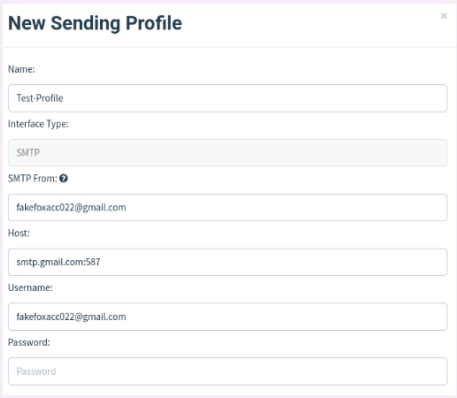


Fig 4.1 — New Sending Profile: name, SMTP host (smtp.gmail.com:587), and sender address filled in

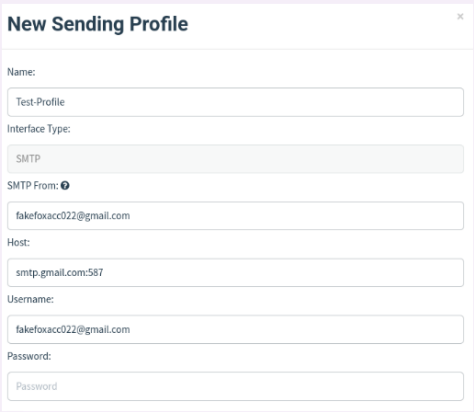


Fig 4.2 — New Sending Profile: all fields entered — ready to add the App Password

4.2 Generating a Gmail App Password

Because Gmail blocks direct SMTP login when 2FA is active, an App Password must be generated. App Passwords are 16-character tokens that allow specific applications (like Gophish) to authenticate to Gmail without using the account's regular password.

Step 7: On the Gmail account, ensure 2-Factor Authentication (2FA) is enabled first.

Step 8: Navigate to: myaccount.google.com/apppasswords — enter a name (e.g., 'Gophish') and click Create.

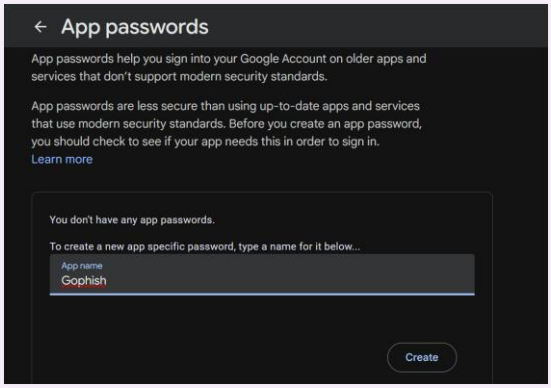


Fig 4.3 — Google Account: App Passwords page — creating a new app password named 'Gophish'

Step 9: Copy the generated 16-character App Password and keep it ready for Gophish.

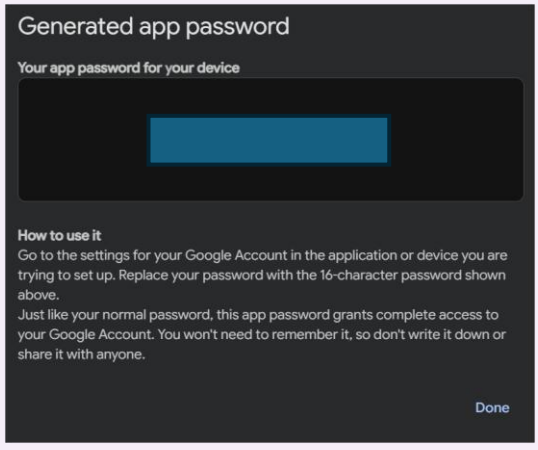


Fig 4.4 — Generated App Password displayed (16 characters) — copy before closing

Note: The App Password is only shown once. Copy it immediately — if it is lost, delete the entry and generate a new one. Never share App Passwords; they grant full Gmail account access.

4.3 Completing the Profile & Sending a Test Email

Paste the App Password into the Password field of the Gophish Sending Profile. Check 'Ignore Certificate Errors' to avoid TLS certificate issues. Then use the Send Test Email button to verify the configuration.

Step 10: Paste the App Password into the Password field. Check 'Ignore Certificate Errors'. Click Send Test Email.

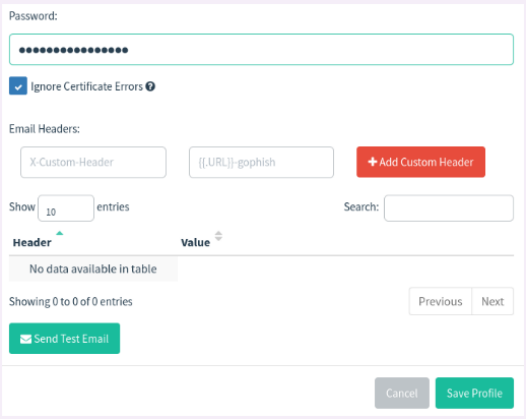


Fig 4.5 — Sending Profile: App Password entered, Ignore Certificate Errors checked, Send Test Email button visible

Step 11: In the Send Test Email dialog, enter a test recipient name and email address, then click Send.

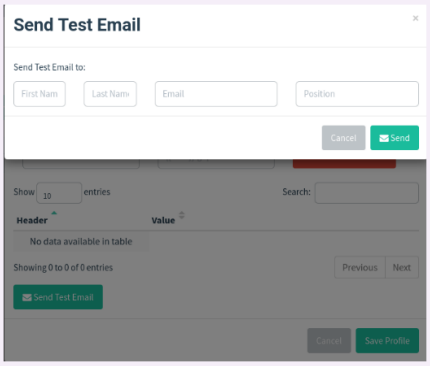


Fig 4.6 — Send Test Email dialog: recipient fields (First Name, Last Name, Email, Position)

Step 12: Confirm the 'Email Sent!' success message appears in the dialog.

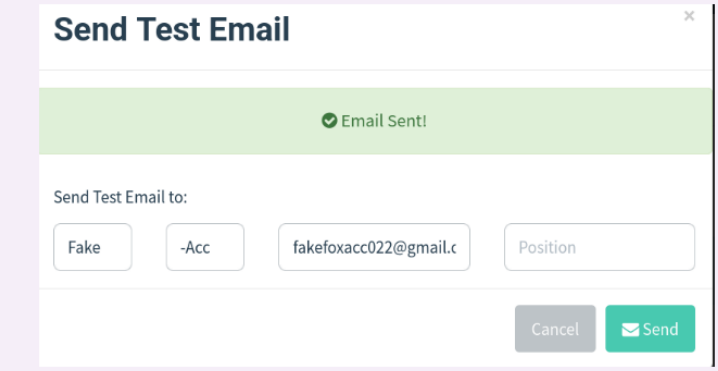


Fig 4.7 — Send Test Email: green 'Email Sent!' confirmation message displayed

4.4 Verifying Email Receipt

The test recipient inbox should show a Gophish test email confirming the sending profile is correctly configured. The email body lists the sending address and recipient details, confirming end-to-end delivery.

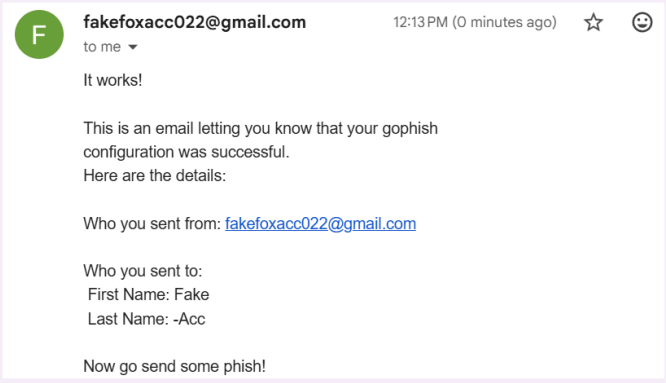


Fig 4.8 — Test email received in inbox: 'It works! Your gophish configuration was successful.'

> **Tip:** The received test email confirms SPF and DKIM are passing for the Gmail account. A delivered test email means the full campaign will reach recipients' inboxes rather than spam folders.

4.5 Saving the Profile

After confirming the test email was delivered, click Save Profile to store the sending configuration. The profile will now appear in the Sending Profiles list and can be selected when creating campaigns.

Step 13: Click Save Profile to save the sending configuration.

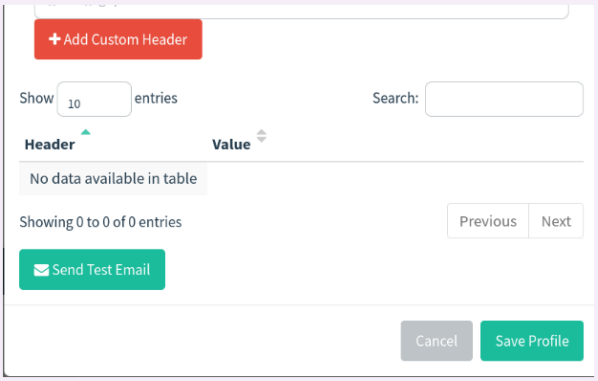


Fig 4.9 — Sending Profile: Save Profile button and profile ready to be saved

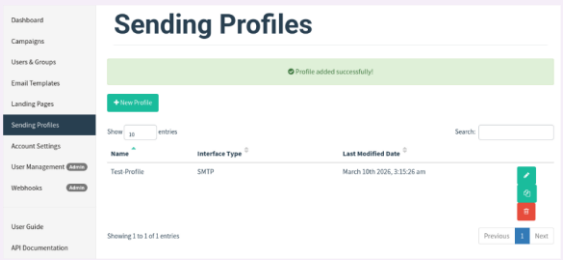


Fig 4.10 — Sending Profiles list: 'Test-Profile' saved successfully (March 10th 2026, 3:15:26 am)

5. Landing Page Setup

A Landing Page is the fake webpage that victims are directed to when they click the link in the phishing email. Gophish can import any live website by URL, inject credential-capturing JavaScript, and optionally redirect the victim to a legitimate page after submission to avoid raising suspicion.

5.1 Creating a New Landing Page

Navigate to Landing Pages in the sidebar and click + New Page. Name the page and use the Import Site button to clone any existing website by URL, or paste custom HTML directly into the editor.

Step 14: Click Landing Pages in the sidebar, then click + New Page.

Step 15: Enter a name (e.g., TOP-SHOP). Click Import Site and enter the target URL to clone, or paste HTML.

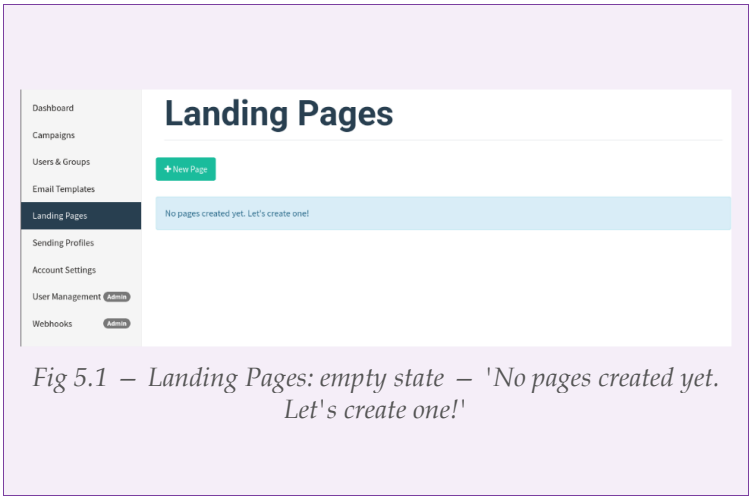


Fig 5.1 — Landing Pages: empty state — 'No pages created yet. Let's create one!'

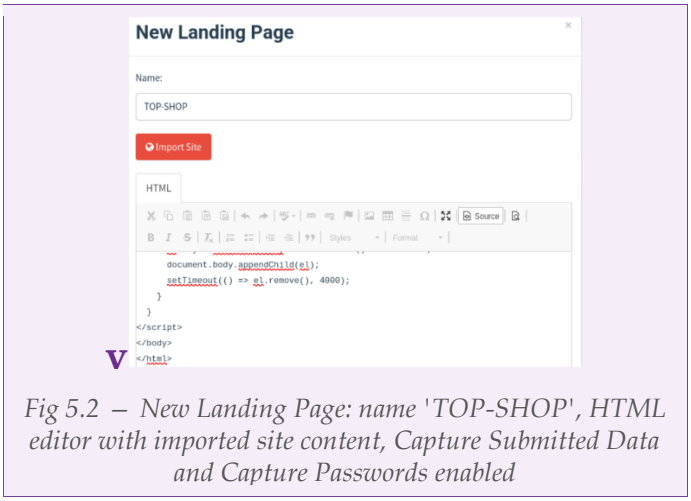


Fig 5.2 — New Landing Page: name 'TOP-SHOP', HTML editor with imported site content, Capture Submitted Data and Capture Passwords enabled

5.2 Enabling Credential Capture & Redirect

Two critical checkboxes must be enabled: 'Capture Submitted Data' records all form fields the victim fills in, and 'Capture Passwords' specifically logs password field values in cleartext. The 'Redirect to' field sets where victims are sent after submitting the form — using a legitimate URL prevents them from realizing they were phished.

Step 16: Check both 'Capture Submitted Data' and 'Capture Passwords'.

Step 17: In the 'Redirect to' field, enter a legitimate URL (e.g., https://google.com) for post-submission redirect.

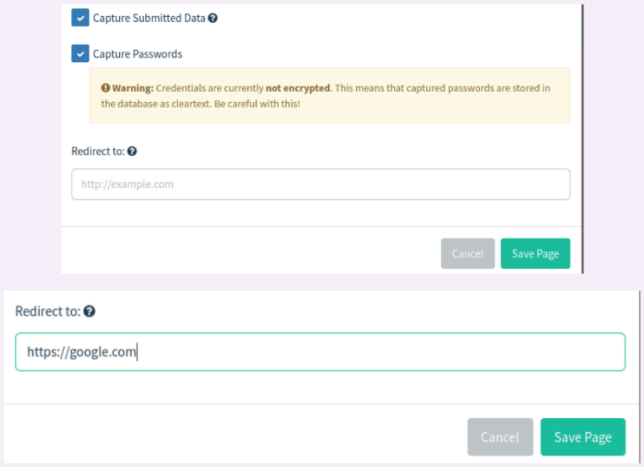


Fig 5.3 — Landing Page: Redirect to field set to https://google.com after credential capture

> **Important:** Gophish warns that captured passwords are stored as cleartext in its database. Treat the Gophish database file as sensitive — restrict access to authorized personnel only.

5.3 Saving the Landing Page

Step 18: Click Save Page. The landing page will appear in the Landing Pages list.

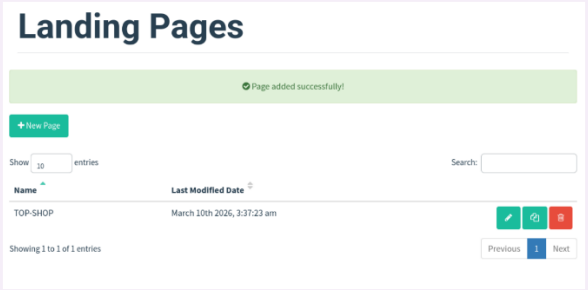


Fig 5.4 — Landing Pages list: 'TOP-SHOP' saved successfully (March 10th 2026, 3:37:23 am)

6. Email Template Creation

An Email Template defines the content of the phishing email that targets receive. Gophish supports importing a real email's raw source — this is particularly effective because the imported HTML preserves the legitimate sender's branding, making the phishing email nearly indistinguishable from a genuine message.

6.1 Obtaining a Real Email for Import

To create a convincing phishing template, a real marketing or notification email from a recognized brand (Smallpdf in this lab) was used. The raw email source was obtained from Gmail by opening the email and selecting 'Show original' from the three-dot menu.

Step 19: Open a legitimate email in Gmail (e.g., Smallpdf welcome email). Click the three-dot menu > Show original.

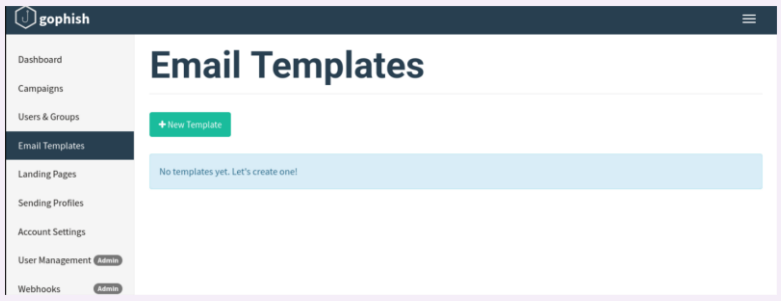


Fig 6.1 — Email Templates: empty state — 'No templates yet. Let's create one!'

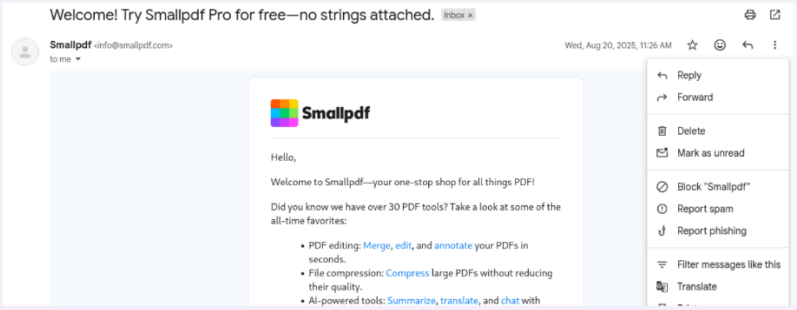


Fig 6.2 — Gmail: legitimate Smallpdf email opened — accessing the three-dot menu for 'Show original'

6.2 Copying the Raw Email Source

The 'Show original' view displays all email metadata including Message-ID, timestamps, SPF/DKIM/DMARC authentication results, and the full raw MIME source. The raw source is copied and pasted into Gophish's Import Email dialog to create the template.



Fig 6.3 — Gmail: Original Message headers — SPF: PASS, DKIM: PASS, DMARC: PASS (from Smallpdf)

Note: Checking that the source email passes SPF, DKIM, and DMARC is useful — it confirms the email structure is valid and the resulting template will look authentic to recipients.

6.3 Importing the Email into Gophish

In Gophish, navigate to Email Templates and click + New Template. Use the Import Email button to open the import dialog. Paste the full raw email source into the Email Content field. The 'Change Links to Point to Landing Page' checkbox automatically rewrites all hyperlinks in the email to point to the Gophish landing page URL.

Step 20: Click Email Templates > + New Template. Click Import Email.

Step 21: Paste the raw email source into the Email Content field. Ensure 'Change Links to Point to Landing Page' is checked. Click Import.

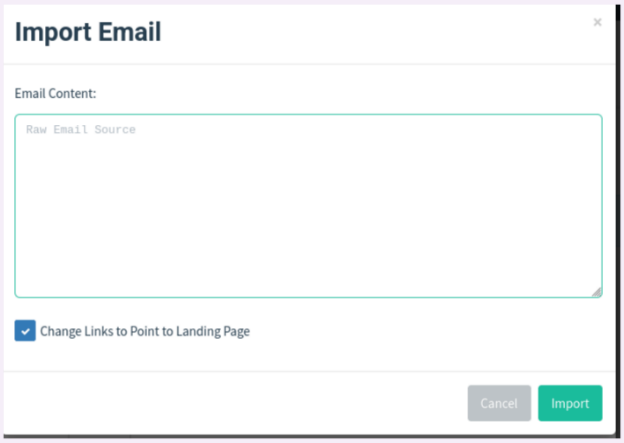


Fig 6.4 — Import Email dialog: empty Raw Email Source field with 'Change Links to Point to Landing Page' checked

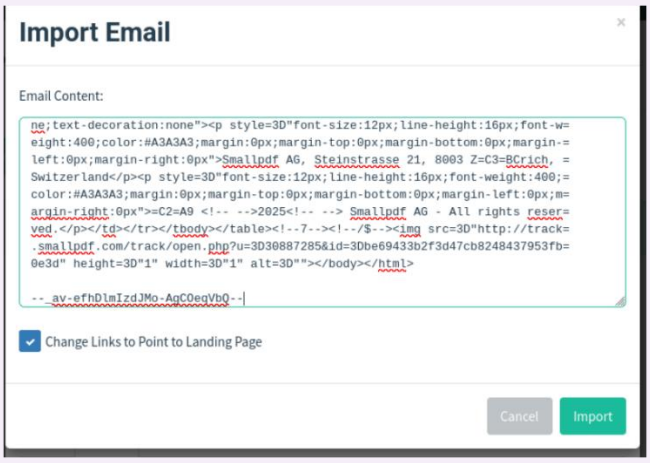


Fig 6.5 — Import Email dialog: full raw HTML of the Smallpdf email pasted into the Email Content field

6.4 Reviewing & Saving the Template

After importing, Gophish renders a preview of the email in the HTML editor. The subject line is auto-populated from the original email. The 'Add Tracking Image' checkbox inserts a 1x1 invisible tracking pixel that logs when a recipient opens the email.

Step 22: Review the rendered email preview. Verify the subject line, sender address, and HTML rendering are correct.

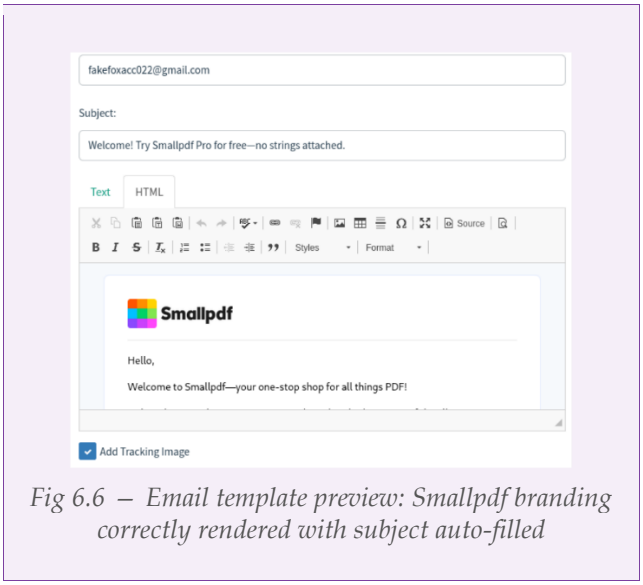


Fig 6.6 — Email template preview: Smallpdf branding correctly rendered with subject auto-filled

Step 23: Ensure 'Add Tracking Image' is checked. Optionally add file attachments with + Add Files. Click Save Template.

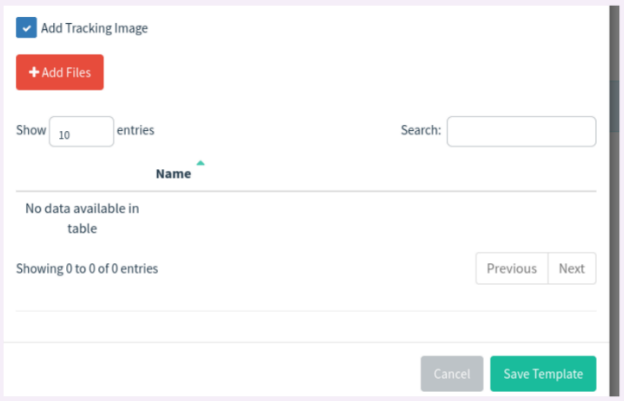


Fig 6.7 — Email template: Add Tracking Image checked, Add Files option, Save Template button

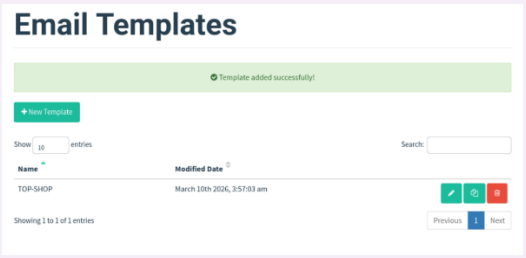


Fig 6.8 — Email Templates list: 'TOP-SHOP' template saved successfully (March 10th 2026, 3:57:03 am)

7. Users & Groups (Target List)

The Users & Groups section defines who receives the phishing campaign. Recipients can be added manually one by one or bulk-imported from a CSV file. Each entry includes first name, last name, email address, and an optional position/role field.

7.1 Creating a New Group

Navigate to Users & Groups in the sidebar and click + New Group. Assign a group name and add recipients individually using the input fields, or upload a CSV using the Bulk Import Users button.

Step 24: Click Users & Groups in the sidebar, then click + New Group.

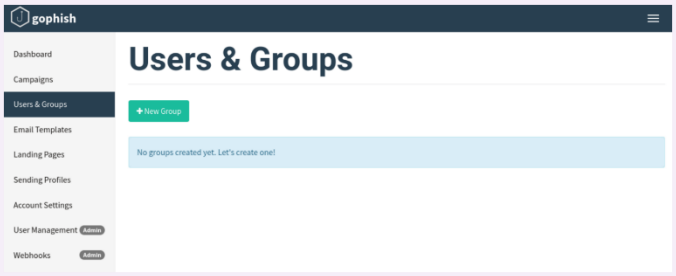


Fig 7.1 — Users & Groups: empty state — 'No groups created yet. Let's create one!'

Step 25: Enter a group name (e.g., TOP-SHOP). Add recipients by filling in First Name, Last Name, Email, and Position fields, then click + Add for each.

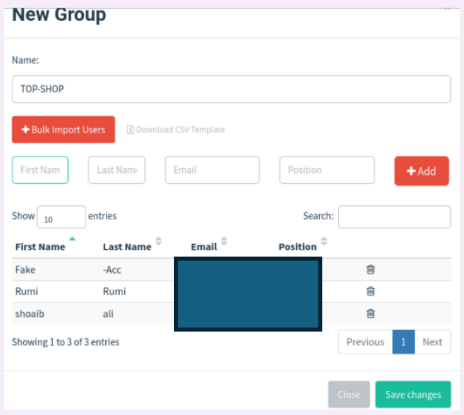


Fig 7.2 — New Group: 'TOP-SHOP' group with 3 recipients added

Step 26: Click Save changes to store the group.

> **Tip:** For large campaigns, use the Download CSV Template button to get the correct column format, populate it with recipients, then use Bulk Import Users to upload all at once.

8. Creating & Launching the Campaign

With all five components configured — Sending Profile, Landing Page, Email Template, and Users & Groups — a Campaign can now be created. The Campaign ties them all together, adds a tracking URL, and schedules the launch.

8.1 Preparing the Campaign URL

The Campaign URL is the link embedded in phishing emails that directs victims to the Gophish landing page. Since the Gophish server runs locally (127.0.0.1), the local URL is shortened using Bitly to create a public-facing link that can be tracked and disguised.

Step 27: Visit bitly.com and shorten the Gophish local server URL (e.g., http://127.0.0.1/) to get a public short link.

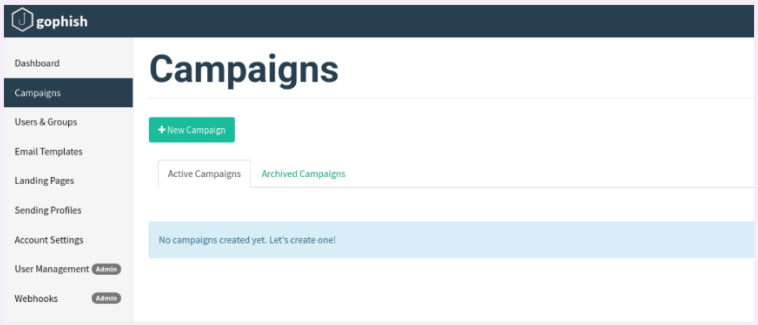


Fig 8.1 — Campaigns: empty state — 'No campaigns created yet. Let's create one!'

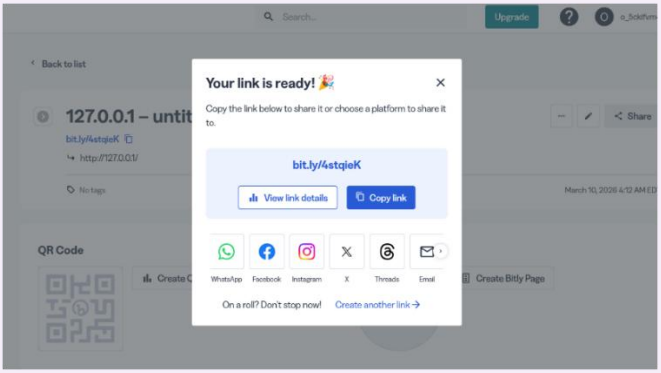


Fig 8.2 — Bitly: shortened link ready — bit.ly/4stqieK pointing to http://127.0.0.1/

Note: In a real engagement, the landing page URL would be hosted on a public-facing server with a registered domain. Using Bitly here demonstrates how the local lab URL is made accessible and disguised for the email link.

8.2 Configuring the Campaign

Click Campaigns in the sidebar, then + New Campaign. The campaign form requires a name, email template, landing page, tracking URL, launch date, sending profile, and target group.

Step 28: Click Campaigns > + New Campaign. Fill in all required fields:

- Campaign Name: a descriptive name for identification.
- Email Template: TOP-SHOP (created in Section 6).
- Landing Page: TOP-SHOP (created in Section 5).
- URL: the Bitly short link (e.g., https://bit.ly/4stqieK) — Gophish uses this in the tracking pixel and links.
- Launch Date: set automatically to current date/time (March 10th 2026, 4:10 am in this lab).
- Sending Profile: Test-Profile (created in Section 4).
- Groups: TOP-SHOP (created in Section 7).

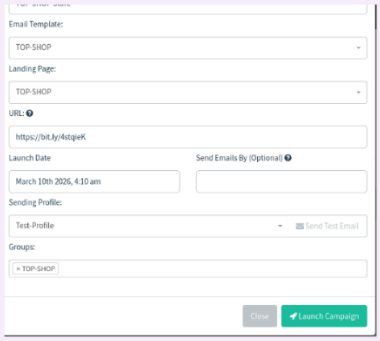


Fig 8.3 — New Campaign form: all fields configured — Template, Landing Page, URL, Sending Profile, Groups all set

8.3 Launching the Campaign

Once all fields are filled in, click Launch Campaign. Gophish presents a confirmation dialog before scheduling the campaign for delivery.

Step 29: Review all campaign settings, then click Launch Campaign.

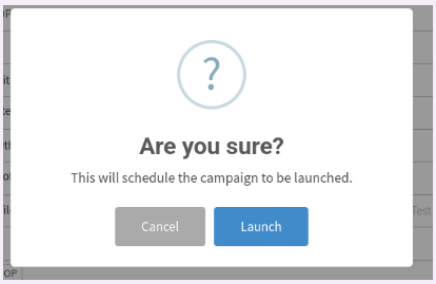


Fig 8.4 — 'Are you sure?' confirmation dialog: 'This will schedule the campaign to be launched.' — Cancel or Launch

Step 30: Click Launch in the confirmation dialog to schedule and send the campaign.

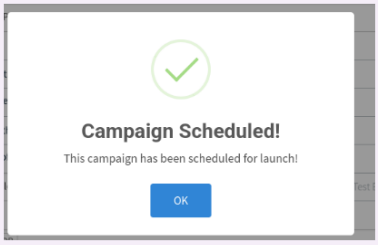


Fig 8.5 — Campaign Scheduled: green checkmark with 'This campaign has been scheduled for launch!'

8.4 Campaign Results & Delivery Status

After launch, Gophish displays the campaign details table showing each recipient's delivery status in real time. Statuses include: Email Sent, Clicked Link, Submitted Data, and Error. In this lab, two recipients showed an Error status due to recipient-side email filtering.

Step 31: Review the Details table to monitor per-recipient delivery status.



Fig 8.6 — Campaign Details: recipients listed with Error status

Note: An Error status in Gophish typically indicates the email was rejected or bounced at the recipient's mail server — not a failure of Gophish itself. This can occur due to recipient-side spam filters, domain reputation, or MX record restrictions. Successful delivery was already verified in Section 4.4 via the test email.

9. Findings Summary

Component / Step	Status	Notes
Installation (apt install)	✓ Success	Gophish v0.12.1 already at newest version on Kali Linux.
First Login & Password Reset	✓ Success	Default creds accepted; password reset to Kali1212 on first login.
Sending Profile — Gmail SMTP	✓ Success	App Password generated; smtp.gmail.com:587 configured correctly.
Test Email Delivery	✓ Success	Test email received — 'It works!' confirmation displayed.
Landing Page (TOP-SHOP)	✓ Success	Site cloned, credential capture + redirect to google.com set.
Email Template (TOP-SHOP)	✓ Success	Smallpdf email imported; tracking image and link rewriting enabled.
Users & Groups (TOP-SHOP)	✓ Success	3 recipients added: Fake-Acc, Rumi Rumi, shoaib ali.
Campaign Launch	✓ Scheduled	Campaign scheduled and confirmed with 'Campaign Scheduled!' dialog.
Campaign Delivery	⚠ Error	Both recipients showed Error status — likely recipient-side spam filtering, not a Gophish fault.

10. Conclusion

This lab successfully demonstrated the complete Gophish campaign workflow from installation to launch. Every component — sending profile, landing page, email template, and target group — was configured and verified. The only partial issue was the campaign email delivery error, which is attributable to recipient-side filtering rather than any misconfiguration of Gophish itself (confirmed by the successful test email in Section 4.4).

Gophish's web dashboard makes it straightforward to create and manage realistic phishing simulations, track engagement metrics (opens, clicks, submissions), and analyze results — making it a valuable tool for both offensive security professionals and defensive security awareness training programs.

10.1 Defensive Recommendations

- Deploy email security gateways with SPF, DKIM, and DMARC enforcement to reject spoofed or unauthorized senders before emails reach inboxes.
- Enable spam and phishing filters at the mail server level — these blocked campaign delivery in this lab, demonstrating their effectiveness.
- Conduct regular phishing simulation exercises using tools like Gophish to train employees to recognize and report suspicious emails.
- Implement URL inspection and sandboxing to analyze links before allowing browser navigation.
- Configure email clients to display full sender addresses and warn when links do not match their display text.
- Monitor and alert on unusual login activity that may indicate credential theft from successful phishing attacks.

> **Important:** All activities in this report were conducted exclusively in a closed, authorized lab environment using test accounts. No real users, production systems, or third-party services were targeted without consent.